PROPRA

PROPERTY INTELLIGENCE PLATFORM

MASTER PRODUCT SPECIFICATION

Version 3.0 · Build-Ready · Confidential & Proprietary

Commission-only · Zero setup fees · Every screen wired to execution

$125M+	$700k+	0%	22+
Managed portfolio (demo)	Avg annual uplift per portfolio	Setup fee — ever	Screens, every decision

1. Purpose of This Document

This is the single source of truth.

It supersedes all previous briefs, demos, and verbal instructions. Every screen, feature, API integration, data source, design decision, and UX pattern is defined here. The team builds exactly what is in this document — no interpretation required.

What failed before — and why this document exists differently

The original brief was excellent but was given to a builder who extracted the data layer and built a display product. Propra is not a dashboard. Propra is an execution engine. Every screen must answer three questions:

- What is the issue and how much is it costing?
- What exactly will Propra do about it?
- What does the owner need to tap to make it happen?

The demo artifact (PropIQ) captured this philosophy correctly. The built version did not. This document rebuilds the specification from first principles, incorporating the original brief, the demo design language, the updated visual direction, and all real-world API integrations.

2. Product Philosophy

The principles that govern every design and build decision

Everyone else does this	Propra does this instead
Shows you the data, you figure out the action	Tells you the action and the exact dollar value of taking it
Charges £5k–£50k/yr before delivering any value	Charges nothing until value is delivered — commission only

Insurance section = a broker referral link	Insurance section = Propra places direct with carriers today
Energy section = "call your broker"	Energy section = Propra switches the contract now
AI = a chatbot on a side panel with no context	AI = pre-loaded at every screen with full live portfolio data
Hold/sell = hire an advisor for £20k	Hold/sell = built-in scenario model with clear recommendation
Compliance = a list of certificates	Compliance = fine exposure in £/$ with one-tap certificate ordering
Rent review = your agent reminds you at renewal	Rent Clock = countdown + tenant leverage score + draft letter ready

UX Principle — every screen follows this exact structure:

1. See it	2. Understand it	3. Do it
The issue is surfaced with live data — not estimated, not manual.	The cost of inaction is quantified in £/$. No jargon. Plain English.	One tap or click. Propra executes — places, switches, books, tenders, drafts.

3. Brand & Design System

Exact specifications — no deviation

3.1 Colour Palette

Colour	Role	Hex	Usage	Notes
Primary Dark	Hero & sidebar	**#0B1622**	Nav, hero cards, dark sections	*Navy-black. Main brand anchor.*
Propra Green	Primary action	**#0A8A4C**	CTAs, positive states, progress	*All "do it" buttons. Active nav.*
Signal Amber	Opportunity	**#F5A94A**	Opportunity badges, warnings	*Money on the table = amber.*
Decision Blue	AI/intelligence	**#1647E8**	AI labels, portfolio score	*Analytics and AI features.*
Alert Red	Risk & urgency	**#D93025**	Compliance risk, urgent items	*Fine exposure, expired certs.*
Background	Page canvas	**#F7F8FA**	Main page background	*Light gray — not white.*

White	Content cards	**#FFFFFF**	All data cards	*Clean white on gray bg.*
Green border	Good status	**#0A8A4C**	Positive metric card borders	*2px left border on cards.*
Amber border	Review needed	**#F5A94A**	Cards needing attention	*2px left border on cards.*
Red border	Action required	**#D93025**	Urgent / non-compliant cards	*2px left border on cards.*

3.2 Typography

Element	Font	Size / Weight	Where used
Hero stat numbers	Instrument Serif	48–72px / 700	Portfolio value, NOI, G2N % on dashboard hero
Section stat numbers	Instrument Serif	32–48px / 600	Card KPIs, opportunity values (£ $43,800/yr)
Card headings	Geist / Arial	16px / 600	Card titles, screen headings
Body text	Geist / Arial	14px / 400	Descriptions, supporting copy
Labels / metadata	Geist / Arial	12px / 400	Timestamps, secondary labels
Navigation	Geist / Arial	13px / 500	Sidebar nav items
CTA buttons	Geist / Arial	14px / 600	Primary action buttons
Data table	Geist Mono	13px / 400	Numbers in tables, rates, codes

3.3 Layout System

- Sidebar: 220px fixed, dark navy (#0B1622), grouped navigation with section labels and badge counts
- Top bar: 64px, white, page title left + search + action buttons right (Export / + Add Property / Run Full Analysis)
- Alert banner: Full width below top bar, amber background for warnings, actionable link right-aligned
- Content area: White cards on #F7F8FA background, 24px padding, 16px gap between cards
- Card borders: 2px left border colour-coded (green = good, amber = review, red = urgent)
- Hero section: Dark navy card at top of dashboard with portfolio greeting, date, Portfolio Value Score widget
- Stat row: 7–8 KPI tiles in a scrollable horizontal row below hero
- Responsive: Desktop-first (1280px+), tablet breakpoint (768px), mobile (390px)

3.4 Component Patterns

- Opportunity card: Category badge (RENT UPLIFT / INSURANCE / REFINANCE etc.) + large £$X,XXX/yr figure + plain-English description + evidence sentence + Quick win badge (if <4 weeks) + primary CTA
- Action item: Icon + title + property name + days remaining + View [section] → link
- KPI tile: Label (small caps) + large number (Instrument Serif) + trend arrow + benchmark note
- Progress bar: Labelled horizontal bar, colour shows status, % and absolute value
- Portfolio score widget: Circular gauge 0–100, sub-scores (Income / Cost / Growth), status label
- Alert banner: Warning icon + bold label + plain sentence + "Review now →" right-aligned

- Table row: Property name + address + metrics + status badge + action button inline

4. Navigation Structure

22 screens across 6 groups — exact sidebar structure

Group	Screen	Route	One-line purpose
OVERVIEW	**Value Dashboard**	`/dashboard`	Portfolio health score, all KPIs, AI opportunity centre, action queue
OVERVIEW	**All Properties**	`/properties`	Portfolio-level view of all assets with per-property G2N, occupancy, alerts
OVERVIEW	**Portfolio Analytics**	`/analytics`	Deep-dive charts: NOI trend, benchmark vs actual, sector breakdown, geography
INCOME ENHANCEMENT	**Rent Optimisation**	`/rent`	Passing rent vs ERV per tenant, market comps, draft renewal letter, engage tenant
INCOME ENHANCEMENT	**Lease Restructuring**	`/leases`	Lease events calendar, break clause tracking, incentive modelling
INCOME ENHANCEMENT	**Ancillary Revenue**	`/ancillary`	AI scan for 5G masts, EV charging, solar, car park, signage, subletting
INCOME ENHANCEMENT	**AI Opportunities**	`/opportunities`	Full ranked list of all AI-identified opportunities with ROI and execution status
COST REDUCTION	**Insurance Audit**	`/insurance`	Live benchmark vs market, carrier comparison, placement workflow, policy docs
COST REDUCTION	**Utility Switching**	`/energy`	Per-property kWh vs benchmark, anomaly detection, tariff switch execution
COST REDUCTION	**CAM Recovery**	`/cam`	Service charge vs lease entitlement, reconciliation statements, recovery actions
COST REDUCTION	**Tax & Compliance**	`/tax`	Rates liability, SDLT, VAT status, CIS compliance, agent commission tracking
ASSET GROWTH	**Hold vs Sell**	`/hold-sell`	Per-asset scenario model: Sell Now / Hold 3yr / Reposition with IRR and net proceeds
ASSET GROWTH	**Acquisitions Scout**	`/acquisitions`	AI screens live listings, scores fit vs portfolio criteria, pipeline tracker
ASSET GROWTH	**Refinance Centre**	`/refinance`	Live loan vs SOFR/LIBOR, lender comparison, DSCR tracker, refi workflow
ASSET GROWTH	**Planning & Dev**	`/planning`	Planning application status, permitted

			development rights, EPC upgrade paths
OPERATIONS	**Rent Clock**	`/rent-clock`	Countdown to every lease event: expiry, break, review — with tenant leverage scores
OPERATIONS	**Tenants**	`/tenants`	Tenant register, payment history, correspondence log, willingness-to-pay score
OPERATIONS	**Work Orders**	`/work-orders`	Maintenance requests, vetted contractor tendering, job tracking, cost vs budget
OPERATIONS	**Compliance**	`/compliance`	Every certificate tracked: EPC, fire, legionella, asbestos, insurance void risk + fine £$
PLATFORM	**Opportunity Inbox**	`/inbox`	Unified action queue ranked by annual value — single place to act on everything
PLATFORM	**Ask Propra AI**	`/ask`	Full-context AI assistant with portfolio data pre-loaded, any question in plain English
PLATFORM	**Settings**	`/settings`	Properties, integrations, notifications, billing, team access, API keys

5. Screen Specifications

Every screen defined: layout, components, data, actions, APIs

5.1 Value Dashboard (/dashboard)

Purpose

The first screen every user sees. Must immediately show: portfolio health score, total value at risk, unactioned opportunity £$, and ranked list of actions. A user should be able to understand the state of their entire portfolio and know exactly what to do next within 10 seconds of loading.

Layout (top to bottom):

- Alert Banner — conditional, amber if any urgent compliance/insurance items. Shows item count + estimated fine/risk exposure + "Review now →"
- Hero Card (dark navy) — "Good morning [Name] — here's your portfolio" + date + portfolio subtitle + Portfolio Value Score widget (circular gauge 0–100 with sub-scores)
- KPI Strip — 7 tiles: Portfolio Value | Gross Monthly Rent | NOI | Occupancy % | Total Sq Ft | Avg NOI Yield | Costs Saved YTD + Unactioned Opportunity tile (amber, always last)
- 3-column analytics row — Geographic Spread (horizontal bar per market) | Asset Class Mix (dot chart) | Top 5 by NOI Yield (ranked list)
- AI Opportunity Centre — heading "Ranked by annual value — every action is executable inside Propra" + 3-column card grid. First card highlighted as HIGHEST ROI. Each card: category badge + £$X,XXX/yr headline + plain-English description + evidence sentence + time estimate + CTA button
- Lease Expiry Profile — stacked bar chart by quarter (Q1–Q8) with urgent/review/secure colour coding
- AI Opportunity Summary — 4 tiles: Income | Cost Saves | Refi/Value | Value Uplift — each with £$ total and action count
- Portfolio Assets table — all properties with G2N, Occ%, ERV uplift badge, quick actions

- Action Required — right column feed of urgent items with property, days, and "View [section] →"

KPI Data Sources:

- Portfolio Value: Sum of per-property estimated values (CoStar AVM or manual override)
- GMR/NOI: Aggregated from lease data + Xero/QuickBooks accounting integration
- Occupancy: Active leases / total lettable area — live from lease register
- NOI Yield: NOI / Portfolio Value — calculated field
- Costs Saved: Running total of Propra-executed savings (insurance + energy + CAM)
- Unactioned Opportunity: Sum of all open AI-identified opportunities not yet actioned

5.2 Insurance Audit (/insurance)

Purpose

Show exactly how much the portfolio is overpaying on insurance vs live market benchmarks, then place replacement policies direct with carriers — no broker. Propra earns 15% of year-1 saving.

Layout:

- Header: "Insurance Premium Audit — AI vs [N] comparable FL commercial policies" + "Rearrange inside Propra →" CTA
- Per-policy table: Policy type | Property/ies | Current annual premium | AI market rate | Saving | Status badge (Overpaying / Competitive / Negligible)
- Total row: "Total identified saving: £$XX,XXX/yr across [N] policies" + "Get quotes inside Propra →"
- Placement workflow (modal/drawer): Carrier comparison grid with live quotes, terms, excess — select + confirm to place
- Placed policies section: Active placements showing carrier, policy number, next renewal, Propra commission earned

Real-world integrations required:

Service	Type	Purpose	Integration method
Lloyd's of London	Carrier API	Commercial property quotes	Direct API — Lloyd's Market Association syndicate access
AIG	Carrier API	Property & casualty quotes	AIG Connect API (REST)
Chubb	Carrier API	High-value property	Chubb API Gateway
Liberty Mutual	Carrier API	Commercial property	Liberty API portal
Travelers	Carrier API	Commercial multi-peril	Travelers Connect API
Aviva (UK)	Carrier API	UK commercial	Aviva Commercial API
CoStar	Market data	Comparable policy benchmarks	CoStar API — insurance claims & risk data
FEMA Flood Maps	Risk data	Flood zone classification per address	FEMA NFHL API (public)
Nuveen / MSCI RCA	Benchmark data	Market insurance rate benchmarks	MSCI Real Capital Analytics API

5.3 Utility Switching (/energy)

Purpose

Per-property energy spend vs live market tariff. Anomaly detection (overnight HVAC, weekend lighting waste). Propra switches the supplier contract — not recommends, switches. Earns 10% of year-1 saving.

Layout:

- Per-utility table: Utility type | Property | Current supplier | Current rate | Market rate | Monthly saving | Annual saving | Status | Action
- Anomaly alerts: AI-detected unusual consumption patterns with cost impact and resolution
- Solar opportunity row: Roof size, estimated generation, FIT/SEISS return, install cost, payback period
- HVAC optimisation: Schedule vs actual runtime, optimised schedule, projected saving
- Total savings footer + "Full energy report →" CTA

Real-world integrations required:

- Octopus Energy Business API — tariff data, contract switching, smart meter data
- EDF Business API — commercial tariff comparison and contract execution
- British Gas Business API — SME commercial tariffs
- Ameresco API — energy efficiency projects and solar feasibility
- E.ON Business API — commercial switching
- US: EIA (Energy Information Administration) API — utility rate benchmarks by state/zip
- UK: Elexon BMRS API — wholesale electricity prices
- EPCI (Energy Performance Certificate) API — current EPC ratings, improvement recommendations
- Smart meter data ingestion: DCC (UK) or Green Button (US) standard for half-hourly reads

5.4 Refinance Centre (/refinance) ← NEW SCREEN

Purpose

Propra monitors every loan in the portfolio daily against live SOFR, LIBOR, and lender rates. When a refinance opportunity exists — rate above market, DSCR improving, LTV headroom available — Propra surfaces it with the annual saving, pre-vetted lender options, and a one-tap "Explore lenders →" workflow.

Layout:

- Header: "Refinance Centre — AI monitors your loans daily vs live SOFR/market rates" + "Compare lenders →"
- Loan table: Property | Current rate | Market rate | Balance | DSCR | Annual saving at refi | Action
- Status logic: Current rate > market rate + 25bps AND DSCR > 1.25 → "Refi now" (green). Within 25bps → "Monitor" (gray). Below DSCR threshold → "Hold" (amber)
- Lender comparison drawer: When "Refi now" clicked — shows pre-vetted lenders, indicative rates, fees, LTV, DSCR requirements. "Request quote" sends Propra-brokered inquiry
- DSCR tracker: Per-property NOI / Debt service — live chart with trend
- LTV calculator: Current valuation / outstanding debt — flags when headroom opens
- Rate alert system: User sets threshold (e.g. "alert me if market drops below 5%") — push notification + email
- Completed refinances: Track closed deals, rate achieved, Propra savings total, commission earned

Real-world integrations required:

Service	Data provided	Integration method
SOFR Index (NY Fed)	Daily SOFR rate	FRED API (Federal Reserve Bank of St Louis) — public, free
ICE Benchmark Administration	LIBOR/Term SOFR rates	ICE Data Services API

Freddie Mac PMMS	Weekly commercial mortgage rates	Freddie Mac Developer API
CoStar / RCA	Cap rates, comparable loan terms	CoStar Capital Markets API
Trepp	CMBS loan performance, market rates	Trepp API (commercial RE debt analytics)
Lender direct APIs	Live indicative rates from pre-vetted panel	Per-lender: PGIM, Blackstone RE Debt, Mesa West, ACORE — direct relationship + API
ATTOM Property Data	AVM valuations for LTV calc	ATTOM API (address-level property data)
CoreLogic	Property valuation, risk data	CoreLogic API (LoanSafe / Property Data)

5.5 Acquisitions Scout (/acquisitions)

Purpose

AI continuously screens live market listings against the portfolio's acquisition criteria (type, geography, cap rate, deal size, asset class). Scores each deal for fit, identifies income optimisation potential, and surfaces it in a ranked pipeline. Propra earns 0.5–1% on completed acquisitions.

Layout:

- Criteria panel: Portfolio type preference, geography, target cap rate range, deal size, asset class weights — editable
- Live pipeline: Ranked deals with AI fit score, address, type, asking price, cap rate, NOI, estimated income uplift, days on market, source
- Deal card expanded: Full AI underwriting summary, comparable sales, income opportunity breakdown, "Request details" / "Track deal" / "Pass" actions
- Closed deals tracker: Acquisition pipeline with offer, due diligence, exchange, completion status
- Market intel panel: CoStar market rent trends, cap rate movement, transaction volume by sector

Real-world integrations required:

- CoStar API — live commercial listings, sales comps, cap rates, rent data, market analytics
- LoopNet API — off-market and listed commercial properties
- Crexi API — commercial real estate listings and analytics
- RealPage / CoStar Market Analytics — rent benchmarks, vacancy, absorption
- Rightmove Commercial API (UK) — UK commercial listings
- EG Group / EG PropertyLink API (UK) — UK investment and occupier market data
- Auction.com API — distressed commercial property pipeline
- ATTOM Data Solutions — ownership records, debt, transaction history
- Anthropic Claude API — deal underwriting narrative, income opportunity analysis, fit score explanation

5.6 Hold vs Sell (/hold-sell)

Purpose

Per-asset scenario modelling: Sell Now vs Hold 3yr vs Reposition. Shows IRR, net proceeds, estimated exit value, income gap. Replaces £20k–£50k of advisory fees per decision. AI provides a clear recommendation with rationale.

Layout:

- Asset selector: Choose property or view all as cards
- Scenario matrix: 3 columns (Sell Now / Hold 3yr / Reposition) × rows (Gross proceeds, Costs, Net proceeds, IRR, Annual yield, Risk rating)
- AI recommendation: Dark card with green/amber/red recommendation label + plain-English rationale + "See full analysis"
- Assumptions panel: Toggle entry price, exit yield assumption, CapEx estimate, rent growth assumption — scenario updates live
- Comparable exit data: Recent sales of similar assets in same market with achieved cap rates
- Tax impact calculator: CGT / SDLT / depreciation recapture estimate per scenario

Real-world integrations required:

- CoStar Capital Markets — comparable sales, cap rate benchmarks, exit yield assumptions
- MSCI Real Capital Analytics — transaction benchmarks by asset class and geography
- ATTOM — AVM valuations, recent sales, ownership history
- IRS/HMRC tax data — CGT rates, SDLT/transfer tax tables (static, updated annually)
- Anthropic Claude API — scenario narrative, recommendation rationale, risk factors

5.7 Compliance (/compliance)

Purpose

Every certificate (EPC, fire safety, legionella, asbestos, contractor insurance, EICR) tracked with expiry countdown. Fine exposure shown in £/$. One-tap certificate ordering from vetted surveyors. Prevents insurance void risk and regulatory fines averaging £8k–£15k per incident.

Layout:

- Summary: Total fine exposure (red number) + certificates expiring <30 days + expired count
- Certificate table: Property | Certificate type | Status | Expiry date | Days remaining | Fine exposure | Action
- Status colours: Green = valid >90d | Amber = expiring <90d | Red = expired or <30d
- Per-item action: "Order now" links to vetted surveyor panel with prices + availability
- Document vault: PDF upload for existing certificates, OCR extraction of expiry date and property
- Reminder system: Auto-email at 90 days, 30 days, 7 days before expiry

Real-world integrations required:

- MHCLG / VOA (UK) — EPC register lookup by address (public API)
- Elmhurst / Stroma Certification — EPC ordering marketplace
- Surveyors network API — RICS-accredited surveyor panel for asbestos, legionella, fire risk, EICR
- Companies House API — contractor insurance verification
- Anthropic Claude API + PDF parsing — OCR and extract expiry dates from uploaded certificates
- AWS Textract or Google Document AI — certificate PDF extraction fallback

5.8 Rent Clock (/rent-clock)

Purpose

Every lease event tracked with a live countdown. Tenant intelligence scores (willingness to pay, move cost, leverage). Propra auto-drafts renewal letters and strategy notes. Prevents rent drift — avg 14% below-ERV gap closed at renewal.

Layout:

- Urgent events: Top section, red cards — lease expiries < 90 days, breaks exercisable now
- Review soon: Amber cards — lease reviews due in 90–365 days
- Secure: Green cards — leases with >1 year until next event

- Each card: Tenant name + property + event type + countdown clock + current rent + ERV + gap % + leverage score + "Draft letter" / "Engage tenant" CTAs
- Leverage score: AI-generated 0–10 score based on tenant sector health, relocation cost, market availability — with explanation
- Renewal strategy: AI-recommended approach (push for open market / offer incentive / fix and extend) with rationale

Real-world integrations required:

- CoStar Tenant Intelligence — tenant business health, sector data, relocation activity
- Companies House API — tenant financial health, credit, director changes
- D&B Hoovers / Creditsafe API — commercial tenant credit scoring
- CoStar / EGi rental comparables — ERV validation by asset type and location
- Anthropic Claude API — leverage score narrative, renewal strategy, draft letter generation

5.9 Ancillary Revenue (/ancillary)

Purpose

AI scans every asset for uncaptured income streams: 5G masts, EV charging, rooftop solar, car park activation, subletting, signage. Quantifies and executes. Avg £124k/yr new income identified. Propra earns 10% of year-1 income.

Real-world integrations required:

- CTIL / Cornerstone / MBNL APIs — 5G mast site suitability scoring and rent benchmarks
- Pod Point / Osprey / bp pulse APIs — EV charging unit economics, installation cost, revenue share
- Ofgem / DNO capacity APIs — grid connection availability per address
- Google Maps Solar API / EA solar potential data — rooftop solar feasibility
- Eversholt / Glenigan — rooftop solar installer marketplace
- APCOA / Q-Park / NCP APIs — car park management and revenue sharing
- JCDecaux / Clear Channel — digital signage/billboard revenue data by location
- Storeroom / Stashbee — subletting income benchmarks

5.10 Planning & Dev (/planning)

Purpose

Planning application status, permitted development rights, EPC upgrade path, and value uplift from change of use or extension. Propra monitors local authority planning portals for decisions affecting portfolio assets.

Real-world integrations required:

- Planning Portal API (UK) — live planning application submissions and decisions
- Local authority APIs (LPA direct) — planning decision notifications per address
- PINS (Planning Inspectorate) API — appeals and call-in decisions
- Historic England / Cadw API — listed building and conservation area data
- OS National Geographic Database — boundary, use class, site area data
- US: Accela Civic Platform API — building permits, planning applications (used by most US cities)
- US: PACER / county recorder APIs — planning and zoning lookups
- Zoneomics / Regrid API (US) — zoning data by parcel
- Anthropic Claude — permitted development analysis, EPC improvement pathway narrative

5.11 Ask Propra AI (/ask)

Purpose

Full-context AI assistant with the entire portfolio pre-loaded. Every screen also has a context-aware AI panel with pre-built prompts specific to that screen. Not a chatbot bolted on — AI is the backbone of the product.

Technical requirements:

- Anthropic Claude API (claude-sonnet-4-20250514) — primary AI model
- Context window must include: all property data, lease register, insurance schedules, energy data, compliance status, open opportunities, recent AI actions
- Per-screen prompt library: Each screen has 4-6 pre-written prompts per screen context. Insurance examples: biggest overpay, draft insurer letter, hurricane exposure explanation
- Free-form: User can ask anything in plain English — "Which tenant is most likely to leave?", "What would happen to my NOI if I refinanced Brickell?"
- Action execution from chat: AI can trigger actions directly. Example: Switch the Wynwood energy contract executes via energy API
- PDF generation: Any AI response can be exported as a branded PDF report
- Memory: Conversation history stored per user/portfolio — AI remembers context across sessions

6. Complete API & Data Integration Map

Every external service Propra connects to — vendor, type, purpose, implementation

6.1 Property Data & Valuations

Service	Category	Market	Data provided + integration notes
CoStar	Listings + analytics	US + UK	Commercial listings, sales comps, rent benchmarks, tenant data, cap rates, market analytics. CoStar API (requires enterprise subscription). Primary source.
LoopNet	Listings	US	Listed and off-market commercial properties. LoopNet API (CoStar subsidiary).
Crexi	Listings	US	CRE marketplace, investment deals. Crexi Platform API.
EG PropertyLink	Listings + data	UK	UK commercial investment and occupier market. EG Group API.
Rightmove Commercial	Listings	UK	UK commercial property listings. Rightmove API (data feed).
ATTOM	Property data	US	Ownership, AVM, transaction history, liens, tax data. ATTOM API (REST/JSON).
CoreLogic	Property data	US	LoanSafe valuations, property risk, ownership. CoreLogic Data API.
MSCI RCA	Transaction data	Global	Investment sales benchmarks by asset class. RCA API (premium).
HM Land Registry	Transactions	UK	Title register, price paid, ownership. HMLR Open Data API (free).
OS National Map	Geographic	UK	Property boundaries, site area, use class. Ordnance Survey API.
Regrid / Zoneomics	Zoning	US	Parcel data, zoning classification. Regrid API.

6.2 Insurance Carriers

- Lloyd's of London — syndicate market access via Lloyd's Market Association API + coverholder arrangement

- AIG — AIG Connect API for commercial property and casualty
- Chubb — Chubb API Gateway for high-value and specialty commercial
- Liberty Mutual — Liberty API portal, SME commercial focus
- Travelers — Travelers Connect, commercial multi-peril
- Aviva (UK) — Aviva Commercial API, UK market
- Allianz — Allianz Global API for large commercial portfolios
- Markel — specialist commercial property, API via managing agent
- Benchmark source: MSCI RCA insurance cost benchmarks + CoStar risk data + internal Propra placement history

6.3 Energy Suppliers

- Octopus Energy Business API — tariff data, smart meter, switching. Best SME API coverage.
- EDF Business — commercial tariff comparison, contract API
- British Gas Business — SME tariffs, switching API
- E.ON Business — commercial tariffs
- Ameresco — energy efficiency and solar project financing API
- US: EIA API (Energy Information Administration) — benchmark tariffs by state (free, public)
- US: Green Button standard — smart meter data ingestion (XML, used by most US utilities)
- UK: DCC (Data Communications Company) — smart meter half-hourly reads
- UK: Elexon BMRS API — wholesale electricity price benchmarks

6.4 Financial & Loan Data

- FRED API (Federal Reserve) — SOFR, Fed Funds Rate, Treasury rates. Free, public.
- ICE Benchmark Administration — Term SOFR, LIBOR transition rates
- Freddie Mac Developer API — weekly PMMS commercial mortgage rates
- Trepp API — CMBS performance, loan-level data, market delinquency rates
- Moody's CRE — credit risk, market benchmarks
- Lender panel (direct): PGIM Real Estate, Blackstone Real Estate Debt, Mesa West, ACORE Capital — Propra arranges direct API/data feed as part of lender relationship
- Xero API — accounting integration (P&L, rent receipts, expense data)
- QuickBooks API — accounting integration (US market)
- Open Banking (UK) — direct bank feed via Plaid or TrueLayer for rent receipt confirmation
- Plaid API (US) — bank account data, rent receipt verification

6.5 AI & Document Processing

- Anthropic Claude API — primary AI model for all intelligence: underwriting, lease analysis, letter drafting, opportunity scoring, recommendation narrative, Ask Propra responses
- OpenAI API — fallback / specialist tasks: document OCR + structured extraction, where fine-tuned models outperform
- AWS Textract — PDF certificate extraction (EPC, fire cert, insurance schedules, lease documents)
- Google Document AI — alternative PDF extraction pipeline, especially for complex/scanned documents
- Adobe PDF Services API — PDF generation for Propra-branded reports and documents

6.6 Compliance & Certificates

- MHCLG EPC Register API — EPC ratings lookup by address (England & Wales, free public API)
- Scottish EPC Register API — Scotland EPC data
- Elmhurst Energy / Stroma Certification — EPC assessment marketplace
- Energy Saving Trust / MCS — renewable energy certificate data
- Surveyors panel (vetted) — RICS-accredited firms for asbestos, legionella, fire risk, EICR. Propra has pre-negotiated rates. Order via Propra API → surveyor booking system.
- Companies House API (free) — contractor company verification, director checks, insurance confirmation
- HSE (UK) — asbestos register guidance; COSHH compliance data

6.7 Planning & Development

- Planning Portal API (UK) — application submissions, decisions, appeal status
- PINS API — planning appeal decisions
- Historic England API — listed building, conservation area, scheduled monument data
- Accela Civic Platform API (US) — building permits and planning applications (deployed by 800+ US municipalities)
- PACER / county recorder public APIs (US) — zoning variances, permit history
- Glenigan (UK) — construction project intelligence, planning pipeline data

6.8 Ancillary Income Sources

- CTIL (Cornerstone + MBNL) — 5G mast site suitability, rent benchmarks, application process
- Wireless Infrastructure Association — mast market data
- Pod Point API / Osprey Charging — EV charger unit economics, grid connection, revenue share models
- Ofgem / Distribution Network Operator APIs — grid capacity by postcode (UK)
- APCOA / Q-Park / NCP — car park management revenue share APIs
- JCDecaux / Clear Channel — digital signage / billboard CPM and location revenue data

6.9 Tenant Intelligence

- Companies House (UK) — free. Director data, accounts, credit events, dissolution risk
- D&B Hoovers — commercial tenant credit scoring, company financials
- Creditsafe API — business credit reports, payment behaviour
- CoStar Tenant Intelligence — lease transaction history, expansion/contraction activity
- LinkedIn / web enrichment (via Clay API or Apollo.io) — tenant business growth signals

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7. Document Ingestion & PDF Extraction

How Propra turns uploaded documents into live structured data

Propra must accept the following document types and extract structured data automatically using AI + OCR. No manual data entry required from the owner.

Document type	Data extracted	Processing method	Validation
Lease document (PDF)	Tenant, property, term start/end, break dates, rent, review mechanism, service charge, permitted use, alienation rights	Claude API + AWS Textract	Cross-check rent vs accounting data; flag anomalies
Insurance schedule (PDF)	Insured properties, sum insured, premium, excess, renewal date, carrier, exclusions	AWS Textract + Claude extraction	Compare to market benchmark immediately on upload
Utility bill (PDF/CSV)	Supplier, account, property, period, kWh, unit rate, standing charge, total	Textract OCR + regex fallback	Auto-benchmark vs EIA/Elexon market rate
EPC certificate (PDF)	Property address, current rating, potential rating, recommendations, valid until	MHCLG API lookup first; Textract if not found	Flag if <3yr to expiry

Fire / Legionella / Asbestos cert (PDF)	Certificate type, property, assessor, date, next review, pass/fail	AWS Textract + Claude classification	Auto-set reminder at 90d and 30d before expiry
Rent schedule (Excel/CSV)	Tenant, unit, passing rent, ERV, review date, lease expiry	CSV parse + Excel read (SheetJS)	ERV auto-populated from CoStar comparable data
Loan / mortgage schedule (PDF)	Property, lender, balance, rate (fixed/floating), margin, maturity, DSCR covenant	Claude extraction + regex for numbers	Immediately compare rate to SOFR/market
P&L / accounts (Xero/QB sync or PDF)	Income by property, expense categories, NOI	Xero/QB API preferred; Textract for manual upload	Reconcile against rent roll; flag gaps

8. Revenue Model

Commission-only — Propra earns when clients earn

Revenue stream	Rate	Trigger	Example
Insurance placement	**15% of yr 1 saving**	Policy placed via Propra	*£18k saving → £2,700 commission*
Energy switching	**10% of yr 1 saving**	Contract switched via Propra	*£52k saving → £5,200 commission*
Cost reduction wins	**8% of annualised saving**	Fix identified + executed	*£139k → £11,120 commission*
New income streams	**10% of first-year income**	Income activated via Propra	*£124k new income → £12,400*
Acquisitions advisory	**0.5–1% of deal value**	Deal completed via Propra	*£4M deal → £20k–£40k*
Contractor tendering	**3% of contract value**	Works instructed via panel	*£250k project → £7,500*
Transaction management	**0.25% of deal value**	Sale/purchase managed	*£4M sale → £10,000*
Refinance facilitation	**0.25% of loan value**	Refi completed via lender panel	*£5M loan → £12,500*

Unit economics — 10-asset portfolio:

£139k/yr	£124k/yr	£263k/yr	~£28k
Cost savings identified	New income activated	Total uplift for client	Propra commission yr 1

9. Build Priorities

What to build first — sequenced by revenue impact and client need

Note on sequencing

These are priority levels, not timelines. The team works all tracks in parallel. Priority 1 items must be complete and production-ready before Priority 2 items go live. Every feature must be fully functional — not a demo placeholder — before it ships.

Priority 1 — Core execution features (highest revenue impact):

- ✓ Insurance placement — live carrier quotes → placement workflow → commission capture
- ✓ Energy switching — live tariff comparison → contract switch → saving confirmation
- ✓ Dashboard redesign — action-first layout per Section 5.1, all KPIs live from real data
- ✓ PDF ingestion — lease, insurance schedule, utility bill extraction (full structured data)
- ✓ Refinance Centre — SOFR-linked daily loan monitoring, lender panel, workflow
- ✓ Compliance tracker — all certificate types, live expiry, fine exposure, ordering workflow

Priority 2 — Intelligence features (client retention and upsell):

- ✓ Rent Clock — lease event countdown, leverage scoring, renewal strategy, draft letters
- ✓ AI Opportunity Centre — full ranked list, all categories, execution from card
- ✓ Acquisitions Scout — CoStar/LoopNet live feed, AI scoring, pipeline tracker
- ✓ Hold vs Sell — scenario model with live comparable data and AI recommendation
- ✓ Ask Propra AI — full-context assistant, per-screen prompts, action execution from chat

Priority 3 — Scale features (growth and platform defensibility):

- ✓ Ancillary Revenue scanner — 5G, EV, solar, car park, signage analysis
- ✓ Xero / QuickBooks live accounting sync
- ✓ Work Orders — vetted contractor tendering, job tracking
- ✓ Planning & Dev — planning portal monitoring, permitted development analysis
- ✓ Mobile-responsive design — iOS and Android browser optimisation
- ✓ PDF report generation — branded per-asset and per-portfolio reports
- ✓ Multi-user access — owner + property manager roles with permission controls

10. Technical Stack

Existing infrastructure to reuse + additions required

10.1 Existing infrastructure (retain and reuse)

- Next.js (App Router) — retain, extend
- Prisma ORM + PostgreSQL — retain, schema evolution only
- Railway deployment — retain, already resolving Node 20 LTS / Prisma compatibility

- Auth (existing) — retain
- TypeScript throughout — retain

10.2 Frontend additions

- Instrument Serif font — Google Fonts, for large stat numbers
- Geist font — Vercel Geist, for body and labels
- Recharts or Nivo — replace any placeholder chart library with production-grade
- TanStack Query (React Query) — server-state management for live data polling
- Zustand — client-state management
- React PDF Viewer — in-app document viewing (lease PDFs, certificates)
- Mapbox GL JS — geographic spread map on dashboard

10.3 Backend additions

- API route handlers for each external integration (see Section 6) — Next.js API routes or separate microservices
- Bull / BullMQ (Redis-backed queue) — background jobs for AI analysis, PDF extraction, daily rate polling
- Redis — caching for market data (CoStar, SOFR, energy rates — refresh daily)
- AWS Textract SDK / Google Document AI SDK — PDF extraction pipeline
- Anthropic SDK — Claude API integration for all AI features
- Webhook receivers — insurance carriers, energy suppliers, accounting systems
- Cron jobs — daily SOFR/rate fetch, weekly CoStar data refresh, nightly compliance expiry check

10.4 Infrastructure additions

- S3 / R2 (Cloudflare) — document storage for uploaded PDFs, generated reports
- SendGrid / Resend — transactional email (compliance alerts, opportunity notifications, renewal reminders)
- Pusher / Ably — real-time dashboard updates (live rate changes, AI analysis completion)
- Sentry — error monitoring, especially for API integration failures
- PostHog — product analytics (track feature usage, opportunity conversion rates)

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11. Recommended AI Tools & Services for Build Team

Tools to help build faster and better

Using AI to build AI-powered software

The team should use AI tools throughout the build. This section recommends specific tools for specific tasks — not just "use ChatGPT" but exactly which tool for which part of the build.

Tool	Use case	Why this one
Claude (Anthropic)	Product reasoning, spec interpretation, complex code architecture, prompt engineering for Propra AI features	Best for long-context reasoning, following complex specs, code generation with architectural understanding
Cursor + Claude Sonnet	Day-to-day coding, refactoring, debugging	Best coding IDE with AI integration — Claude Sonnet 3.5/4 for code is exceptional for Next.js/TypeScript
v0 by Vercel	UI component generation	Generates production-ready React/Tailwind from descriptions — perfect for rapid card/screen prototyping

Figma + FigJam AI	Design system, screen mockups, component specs	Once spec is confirmed, design each screen in Figma before building — prevents rework
GitHub Copilot	Inline code suggestions within IDE	Good for boilerplate, API calls, test generation
Perplexity / Exa	API documentation lookup, integration research	Better than Google for finding specific API docs quickly
PostHog AI features	Analytics + session replay	Understand how users actually use the product — feeds back into prioritisation
Linear	Sprint and issue management	Build each screen spec into a Linear ticket with acceptance criteria from this document

Build process recommendation:

- For each screen: Start with this spec → generate component in v0 → refine in Cursor → wire to real API → test with live data → QA against spec acceptance criteria
- For each API integration: Use Perplexity to find exact API docs → use Claude to generate the Next.js API route and types → test in isolation → integrate into screen
- For AI features: Write the prompt in Claude.ai first, iterate until the output is exactly right → then implement as a Claude API call in code
- For PDF extraction: Test each document type manually in AWS Textract console first, understand the output structure → then build the extraction pipeline
- For design: Provide this spec + the demo artifact URL to v0 or Cursor — "build this screen to match the design language in this spec, referencing this demo as the visual target"

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12. Acceptance Criteria

How to know when each feature is done — not just built

Definition of done

A feature is not done when the code works in isolation. It is done when: (1) it shows real data from the correct source, (2) the action it enables actually executes end-to-end, (3) the output matches the design spec exactly, and (4) it has been tested on a real portfolio with real documents.

Insurance placement — done when:

- ✓ Upload a real insurance schedule PDF → system extracts all policy details correctly
- ✓ System benchmarks against live carrier APIs and shows real saving (not mocked data)
- ✓ User can select a replacement policy, confirm, and a real placement instruction is sent to the carrier
- ✓ Commission is recorded in the Propra ledger on placement confirmation

Refinance Centre — done when:

- ✓ Daily SOFR rate is fetched from FRED API and reflected in the rate comparison table
- ✓ Loan data entered manually or extracted from uploaded document shows correct DSCR
- ✓ Clicking "Explore lenders →" shows real lender panel with indicative rates
- ✓ Rate alert emails are sent when threshold is crossed

Dashboard — done when:

- ✓ All KPI tiles show real data from lease register + accounting integration, not demo placeholders
- ✓ AI Opportunity Centre shows real identified opportunities ranked by annual value
- ✓ Portfolio Value Score reflects actual portfolio metrics, not a hardcoded number
- ✓ A new user can log in, upload their lease data and a utility bill, and within 5 minutes see real personalised intelligence on their portfolio

Ask Propra AI — done when:

- ✓ AI has access to the user's real portfolio data in its context window
- ✓ "What is my biggest insurance overpay?" returns the correct answer from real data
- ✓ "Draft a renewal letter for [tenant name]" produces a complete, accurate, usable letter
- ✓ AI can trigger an action ("switch the Wynwood energy contract") and the switch executes

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PROPRA · PROPERTY INTELLIGENCE

Commission-only · No setup fees · Propra fixes it

This document is the spec. Build exactly this.